EE-M-002206(Army-GL)
September 28, 1984

INTERIM FEDERAL SPECIFICATION

MACARONI SALAD, THERMOSTABILIZED, TRAY PACK

This Interim Federal Specification was developed by the U.S. Army Natick Research and Development Center, Natick, MA 01760-5014, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE

1.1 Scope. This specification covers macaroni salad thermostabilized in tray pack cans for use by the Department of Defense as a component of operational rations.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

TT-C-495	- Coatings, Exterior, For Tinned Food Cans
PPP-B-636	- Boxes, Shipping Fiberboard

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

(Single copies of this specification, other Federal specifications, and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General

FSC 8940

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:44Z
Check the source to verify that this is the current version before use.

Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Forth Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.

(Federal Government activities may obtain copies of Federal Standardization Documents, and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

Military Specifications:

MIL-L-1497 - Labeling of Metal Cans for Subsistence Items
MIL-L-35078 - Loads, Unit: Preparation of Nonperishable Subsistence; General Specification for

Military Standards:

MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-900 - Bacterial Standards for Starches, Flours, Cereals, Alimentary Pastes, Dry Milks and Sugars Used in the Preparation of Canned Foods for the Armed Forces

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Laws and Regulations:

U.S. Department of Health and Human Services

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder (21 CFR Parts 1-199)

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:44Z
Check the source to verify that this is the current version before use.

U.S. Department of Agriculture (USDA)

Regulations Governing Inspection and Certification of Processed Fruits and Vegetables and Related Products

United States Standards for Grades of Onions (Other than Bermuda - Granex - Grano and Creole Types)

United States Standards for Grades of Pickles

United States Standards for Grades of Sweet Peppers

(Application for copies should be addressed to the Director, Fruit and Vegetable Division, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

U.S. Standards for Condition of Food Containers

(Application for copies should be addressed to the Director, Market Research and Development Division, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

Environmental Protection Agency (EPA)

National Interim Primary Drinking Water Regulations

(Application for copies should be addressed to the Office of Drinking Water WH550, Environmental Protection Agency, Washington, DC 20460.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

Association of Official Analytical Chemists (AOAC)

Official Methods of Analysis of the Association of Official Analytical Chemists

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:44Z
Check the source to verify that this is the current version before use.

(Application for copies should be addressed to the Association of Official Analytical Chemists, 1111 North 19th Street, Suite 210, Arlington, VA 22209.)

American Oil Chemists Society

Official and Tentative Methods of the American Oil Chemists Society

(Application for copies should be addressed to the American Chemists Society, 508 South Sixth Street, Champaign, IL 61820.)

National Academy of Sciences

Food Chemicals Codex

(Application for copies should be addressed to the National Academy Press, 2101 Constitution Avenue, Washington, DC 20418.)

American Dehydrated Onion and Garlic Association (ADOGA)

Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products

(Application for copies should be addressed to the American Dehydrated Onion and Garlic Association, 251 Kearney Street, No. 410, San Francisco, CA 94108.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.3, 6.1, and 6.3).

3.2 Ingredients. All ingredients shall be clean, sound, wholesome and shall be as specified herein or in referenced documents.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:44Z
Check the source to verify that this is the current version before use.

3.2.1 Macaroni. Elbow macaroni shall be made with enriched durum semolina. The outer curvitive length shall range between 0.035-0.04 inch. The macaroni shall contain glycerol monostearate. Moisture shall not exceed 12 percent.

3.2.2 Salad dressing. The salad dressing shall conform to the Food, Drug, and Cosmetic Act, Standard of Identity for Salad Dressing (21 CFR 169. 150). The salad dressing shall have a pleasing flavor and be free of rancidity or other objectionable flavors or odors. The color shall be white to ivory (off white). The body shall be semi-solid of smooth uniform consistency, free from lumps, watery and/or oil separation.

3.2.3 Celery, chopped, freeze-dried. The product shall consist of celery which has been washed, trimmed, transversely cut into 1/8 to 3/8 inch slices, (with no leaf cuts) sulfited, freeze-dried or partially air dried, and freeze dried. The celery shall have an odor and flavor typical of fresh celery. The product shall reconstitute to a pale to dark green color. The moisture content shall be not greater than 4 percent and the sulfite level shall be in the range of 500 to 1500 ppm as SO_2.

3.2.4 Onion, dehydrated, chopped. The dehydrated chopped onion shall be fancy grade of the Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products.

3.2.5 Peppers, green, sweet, freeze-dried. The peppers shall be 1/4-inch dices of sweet bell peppers which have been sulfited and dehydrated by freeze-drying or partially air dried, followed by freeze drying. The dices shall be free of seeds, seed cells and diseased or rotted areas. The pepper dices shall have a fresh, clean typical sweet green pepper flavor and odor and a green with minimal brownish color. The moisture content shall be not greater than 4 percent and the sulfite level shall be in the range of 500 to 1500 ppm as SO_2.

3.2.6 Peppers, red, sweet, freeze dried. The peppers shall be 3/8-inch dices of sweet, red bell peppers which have been sulfited and dehydrated by freeze-drying or partially air dried following by freeze-drying. The dices shall be free of seeds, seed cells, and diseased or rotted areas. The pepper dices shall have a fresh, clean, typical sweet red pepper flavor and odor and shall reconstitute to a bright orange-red to dark red in color. The moisture content shall be in the range of 5.5 to 7.5 percent and the sulfite level shall be in the range of 500 to 1500 ppm as SO_2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:44Z
Check the source to verify that this is the current version before use.

3.2.7 Vinegar, cider. The cider vinegar shall be prepared from apple cider and shall be clear, light amber in color, free from haze, sediment, suspended matter and floating particles. The cider vinegar shall contain not less than 3.0 nor more than 5.0 percent acidity (acetic acid).

3.2.8 Juice, lemon. Lemon juice shall be prepared from unfermented juice obtained from fresh sound mature and thoroughly cleaned fruit of one or more of the high acid varieties of the species Citrus limon (limonia). Such lemon juice may be canned or frozen, single strength or concentrated.

3.2.9 Salt. Salt shall be non-iodized, white, refined sodium chloride with or without anti-caking agents.

3.2.10 Pepper, white, ground. Ground white pepper shall be derived from the dried mature berries of Piper nigrum L. from which the outer covering or outer and inner coverings have been removed. The ground white pepper shall have a characteristic penetrating odor, a hot biting pungent flavor and a light color. The white pepper shall contain not less than 1.0 mL volatile oil per 100 grams and be of such size that not less than 95 percent shall pass through a U.S. Standard No. 40 sieve.

3.2.11 Garlic powder. Garlic powder shall be fancy grade of the Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products.

3.2.12 Pickles, sweet, whole style. Pickles shall be sweet whole type and shall meet the requirements of U.S. Grade B, or better, of the U.S. Standards for Grades of Pickles.

3.2.13 Pickle juice. Pickle juice shall be drained from sweet pickles that meet the requirements of Grade B, or better, of the U.S. Standards for Grades of Pickles.

3.2.14 Xanthan gum. Xanthan gum shall comply with the requirements of the Food Chemicals Codex.

3.2.15 Starch, waxy maize, modified. The starch shall be white, odorless, finely pulverized, modified waxy maize starch for use in canned foods. The modified starch shall demonstrate initial viscosity development in the

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:44Z
Check the source to verify that this is the current version before use.

temperature range of 140° to 170°F and a typical viscosity (be fully hydrated) at common retort temperatures. The starch shall resist breakdown at low pH, under shear stress, and under conditions of cold storage. The starch shall be bland with essentially no cereal or starch taste and shall comply with MIL-STD-900.

3.2.16 Water. Water used for formulation, ice making and washing shall conform to the National Interim Primary Drinking Water Regulations.

3.2.17 Preblended spice and seasoning mixture. Preblended spices and seasonings may be used. The spices and seasonings in the mixture shall comply with the requirements of this specification. The containers used for the spice and seasoning blend shall be labeled with each ingredient and the percentage of each ingredient in the blend. The ingredients shall be in the same proportions as specified in the ingredient formula.

3.3 Preparation and processing.

3.3.1 Macaroni preparation. The macaroni shall be blanched in boiling water containing 1 ounce of vegetable oil for each 10 gallons of blanch water until the weight increases by two times. The macaroni shall be immediately cooled to less than 90°F and rinsed in cold water to remove excess starch. If the macaroni is not immediately filled it shall be cooled rapidly and held in the range of 30° to 40°F for not longer than 4 hours prior to filling.

3.3.2 Celery preparation. The dehydrated celery slices shall be rehydrated in an excess of cold water for 10 minutes or until a tender texture is achieved. Drain thoroughly before filling. If the rehydrated celery is not immediately filled, it shall be immediately cooled to a temperature of 30° to 40°F and held in this range for not longer than 4 hours prior to filling.

3.3.3 Onion preparation. The dehydrated chopped onion shall be rehydrated in an excess of cold water for 10 minutes or until a tender texture is achieved. Drain thoroughly before filling. If the rehydrated onion is not immediately filled, it shall be immediately cooled to a temperature of 30° to 40°F and held in this temperature range for not longer than 4 hours prior to filling.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:44Z
Check the source to verify that this is the current version before use.

3.3.4 Red pepper preparation. The dehydrated red peppers shall be rehydrated in an excess of cold water for 30 minutes or until a tender texture is achieved. Drain thoroughly before filling. If the rehydrated red pepper is not immediately filled, it shall be immediately cooled to a temperature of 30° to 40°F and held in this temperature range for not longer than 4 hours prior to filling.

3.3.5 Pickle preparation. The sweet pickles shall be drained and chopped into approximately 1/4-inch pieces. Reserve the pickle juice for use in the sauce formulation.

3.3.6 Green pepper preparation. The dehydrated green peppers shall be rehydrated in an excess of cold water for 30 minutes or until a tender texture is achieved. Drain thoroughly before filling. If the rehydrated green pepper is not immediately filled, it shall be immediately cooled to a temperature of 30° to 40°F and held in this temperature range for not longer than 4 hours prior to filling.

3.3.7 Macaroni salad preparation. The macaroni salad shall be prepared as follows:

a. Macaroni and vegetable mixture. The blanched macaroni and rehydrated vegetables shall be combined in the following amounts:

Ingredient	Percent
Macaroni, blanched	50.0
Celery, rehydrated	18.0
Pepper, green, rehydrated	16.0
Onion, chopped, rehydrated	8.0
Pepper, red, rehydrated	4.0
Pickle, sweet	4.0

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:44Z
Check the source to verify that this is the current version before use.

b. Sauce formulation. The sauce shall be prepared from the following ingredients:

Ingredient	Percent
Water	55.73 1/
Salad dressing	34.00
Salt	3.40
Cider vinegar	3.00 1/
Pickle juice	2.00
Lemon juice	1.00
Starch, modified	0.50
Xanthan gum	0.20
Garlic powder	0.09
White pepper	0.08

1/ The percentage of water and vinegar shall be adjusted appropriately to achieve the pH as shown in the finished product requirements.

c. Sauce preparation. The sauce shall be prepared as follows:

1. The water , salt, garlic powder and white pepper shall be placed in a steam kettle. Blend well and heat to 180°F.

2. Prepare a slurry with the vinegar, lemon juice, pickle juice, starch and xanthan gum using a high speed mixer. Add to the kettle and heat to 180°F.

3. Add the salad dressing to the kettle and blend the mixture until smooth.

4. Adjust the sauce weight with water to compensate for evaporative losses.

5. The sauce shall be filled immediately following preparation.

3.4 Tray pack filling and sealing. Each tray pack can (see 5.1.1) shall be filled with 54 ± 1 ounces of macaroni - vegetable mixture and sufficient sauce such as to conform to the finished product requirements and to the following requirements:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:44Z
Check the source to verify that this is the current version before use.

a. The macaroni-vegetable mixture and the sauce shall be filled into the tray in a manner than ensures uniform distribution of the ingredients.

b. Each can shall be hermetically sealed with a minimum of 10 inches (Hg) vacuum immediately after filling.

3.5 Tray pack thermoprocessing. The filled and sealed tray pack cans shall be thermostabilized by retorting at a temperature of 212° to 220°F until a product temperature of 195°F has been achieved.

3.6 Finished product requirements. The finished product shall comply with the following requirements:

a. There shall be no foreign material, such as but not limited to dirt, insect, insect parts, hair, wood, paper, paint, glass, or metal.

b. There shall be no foreign odor or flavor, such as but not limited to burnt, scorched, stale, sour, rancid, musty, or moldy.

c. Product shall be a uniform distribution of off-white macaroni, and naturally colored celery, green and red pepper, onion, and pickles in a slightly thick off-white creamy sauce.

d. The average net weight shall be not less than 106 ounces.

e. No individual can shall contain less than 104 ounces of product.

f. The salt content shall be not less than 1.0 percent nor greater than 1.5 percent.

h. Texture of the macaroni and vegetables shall not be mushy, hard, fibrous or tough.

i. The average pH shall be not less than 4.15 nor greater than 4.45.

j. No individual pH value shall be less than 4.10 nor greater than 4.50.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:44Z
Check the source to verify that this is the current version before use.

3.6.1 Palatability. The finished product shall be equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance.

3.7 Plant qualification. The product shall be prepared, processed, and packaged under sanitary conditions in accordance with good manufacturing practice. The product shall be processed in establishments meeting the sanitary requirements of the applicable government inspection agency.

3.8 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all requirements of this specification. The contractor shall assure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and acceptance service. Product acceptability shall be determined by the USDA. The USDA will determine the degree of acceptance service necessary to assure compliance with the requirements of this specification. The cost of grading and acceptance services performed by the USDA involving inspection, official documentation, and related services shall be borne by the contractor.

4.3 First article inspection. When a first article is required (see 6.1), it shall be inspected in accordance with the quality assurance provisions of this specification and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this specification or any appearance or palatability failure shall be cause for rejection of the first article.

4.4 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards, unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.1 Ingredient and component examination. Conformance of ingredients and components to identity, condition, and other requirements shall be

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:44Z
Check the source to verify that this is the current version before use.

determined by examination of pertinent labels, markings, U.S. Grade certificates, invoices, certificates of analysis, or other valid documents. In addition, each ingredient shall be examined organoleptically as necessary to determine conformance to the condition requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient or component lot or of any involved product.

4.5 In-process examination. In-process examination shall be performed to determine conformance to the preparation, processing, can interior coating, filling, sealing, and packing requirements. Any nonconformance revealed by actual examination or by review of records of time, temperature, and formulation or of other valid documents shall be cause for rejection of the involved product.

4.6 Tray pack inspection. The inspection lot shall include only tray packs produced in one workshift. The USDA reserves the right to separate the inspection lot into smaller inspection lots.

4.6.1 Net weight inspection. Randomly select 30 filled and sealed tray pack cans from the inspection lot and weigh separately. Subtract the average tare weight (determined by randomly selecting and weighing 30 of the empty tray pack cans and lids used in preparing the product and dividing the total weight by 30) from the weight of each tray pack in the sample. Results shall be reported to the nearest 1 ounce. If the average net weight is less than 106 ounces or if the net weight of any individual can is less than 104 ounces, the lot shall be rejected.

4.6.2 Product inspection. The sample size shall be as indicated by the double sampling plan specified in table I. The tray pack cans shall be selected at random from the lot. The tray pack cans shall be opened and inspected for the defects listed in table II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:44Z
Check the source to verify that this is the current version before use.

TABLE I. Double sampling plan for product inspection 1/

Lot size (cans)	Sample size (No. of cans)	Cumulative sample	Acceptance number	Rejection number
0-3200	8	-	0	2
	8	16	1	2
3201-35000	13	-	0	3
	13	26	3	4

1/ a. If no defects are found in the first sample, the lot shall be accepted.

b. If the number of defects found in the first sample equals or exceeds the rejection number, the lot shall be rejected.

c. If the number of defects found in the first sample exceeds the acceptance number but is less than the rejection number, a second sample shall be inspected. Defects found in the first and second samples shall be combined and if the number of defects in the cumulative sample equals or exceeds the rejection number, the lot shall be rejected.

TABLE II. Product inspection 1/ 2/

Category	Defect
Major	
151	Texture of macaroni or vegetables hard, mushy, fibrous or tough.
152	Individual components not uniformly distributed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:44Z
Check the source to verify that this is the current version before use.

TABLE II. Product inspection 1/ 2/ (cont'd)

Category	Defect
Major	
153	Color not characteristic of component ingredients (materially darkened, scorched or oxidized).

1/ The presence of any foreign material (e.g., glass, dirt, insect parts, hair, wood, metal, etc), foreign odor of flavor (e.g., burnt, scorched, moldy, rancid, sour, stale, etc.), or foreign color shall be cause for rejection of the lot.

2/ Product not equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance shall be cause for rejection of the lot. (This comparison shall be performed only when deemed necessary by an AMS agent.)

4.6.3 pH testing. Three tray packs shall be randomly drawn from each lot. For each sample tray, three aliquots shall be randomly drawn from different parts of the tray and the pH determined in accordance with the Cereal Food Chapter of the Official Methods of Analysis of the Association of Official Analytical Chemists. The results shall be reported to the nearest 0.01. The average of these three determinations will constitute the pH of the unit sample. The average of the lot shall be calculated using these unit sample determinations. Acceptance of the lot shall be based on the lot average pH being between 4.15 and 4.45 and no unit sample pH used in determining the lot average being less than 4.10 or greater than 4.50.

4.6.4 Salt content testing. Nine tray pack cans shall be selected at random from the lot and distributed as follows:

- Three for laboratory analysis.
- Three for submission to the contractor.
- Three for retention by the AMS agent as a reserve sample.

The three tray packs for laboratory analyses shall individually be tested for salt content in accordance with the Official Methods of Analysis of the Association of Official Analytical Chemists, except that preparation of the samples shall be as follows: The cans shall be opened and the entire contents

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:44Z
Check the source to verify that this is the current version before use.

of each can shall be separately blended in a Waring blender or equivalent. The lot shall be rejected if the salt content of any individual tray pack is less than 1.0 or greater than 1.5 percent. Analysis of reserve samples at the request of the contractor shall not be permitted, unless the original laboratory analysis indicated that the involved lot will be rejected because of noncompliance with the salt content requirement. When the reserve sample is analyzed, the analyses for salt shall be made and will be considered final. Unused reserve samples shall be returned to the contractor for inclusion in subsequent lots.

4.7 Can condition examination. Examination of filled and sealed tray pack cans shall be in accordance with the United States Standards for Condition of Food Containers except that inspection for labeling shall be in accordance with MIL-L-1497 (see 5.4).

4.8 Shipping container examination. Shipping containers shall be examined for defects in assembly, closure, and reinforcement (when applicable), in accordance with the appendix of PPP-B-636. In addition, the following defects shall be classified as follows: Major: National stock number, item description, contract number, or date of pack markings missing, incorrect, or illegible. Minor: (1) Other required markings missing, incorrect, or illegible; (2) Interior packing with fiberboard liner or pads not as specified; (3) Reinforcement with nonmetallic strapping or tape is not used; (4) Height of box liner less than inside depth dimension of box by more than 1/8 inch or extends above top edge of box; (5) Length or width of pads less than box dimensions by more than 1/8 inch; (6) Arrangement or number of cans not as specified.

4.9 Unit load inspection. Inspection of unit loads shall be in accordance with the quality assurance provisions of MIL-L-35078.

5. PREPARATION FOR DELIVERY

5.1 Packaging. The product shall be packaged in accordance with level A or C as specified (see 6.1).

5.1.1 Level A. A size 1001 by 1206 by 200 tray pack can shall be filled with 106 ounces of the food product, sealed, and thermoprocessed as specified in 3.4, 3.5, and 3.6. The base plate of the can shall be electrolytic chromium coated steel and shall be of sufficient temper to protect the product during shipment and storage. The base plate weight of the body and end of each can shall be 90 pounds per base box. The entire inside area of the can shall be coated. Scratches or fractures shall not penetrate through the interior can coating. The interior can coating shall not peel or blister when in contact with

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:44Z
Check the source to verify that this is the current version before use.

the product. The interior can coating shall neither affect nor be affected by the packaged product. A certificate of conformance shall be furnished to certify compliance with the interior can coating requirements. The can shall be coated overall on the outside with a coating conforming to type I of TT-C-495.

5.1.2 Level C. The product shall be packaged as specified in 5.1.1, except that cans with commercial exterior coating will be acceptable.

5.2 Packing. The product shall be packed in accordance with level A, B, or C, as specified (see 6.1).

5.2.1 Level A packing. Four cans of product, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard box, constructed and closed in accordance with style RSC-L, grade V2s of PPP-B-636. The cans shall be packed flat, four in depth within the box. The inside of each box shall be provided with a box liner and five fiberboard pads. The height of the box liner shall be equal to the full inside depth of the box (+0 inch, -1/8 inch). The pads shall be not less than 1/8 inch of the full length and width dimensions of the box and shall be fabricated of grade V3c fiberboard. The pads shall be placed between the cans and on the top and bottom of the stacked cans. Each fiberboard box shall be reinforced with nonmetallic strapping or pressure-sensitive adhesive filament reinforced tape in accordance with the appendix of PPP-B-636. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.1). Strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.2.2 Level B packing. Four cans of product, packaged as specified in 5.1, shall be packed as specified in 5.2.1, except the box shall be constructed of grade V3c, V3s, or V4s fiberboard.

5.2.3 Level C packing. Four cans of product, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard box, constructed and closed in accordance with style RSC-L, class domestic, grade 275 of PPP-B-636. The cans shall be packed flat, four in depth within the box. The inside of each box shall be provided with a box liner and five fiberboard pads. The height of the box liner shall be equal to the full inside depth of the box (+0 inch, -1/8 inch). The pads shall be placed between the cans and on the top and bottom of the stacked cans. The pad dimensions shall be not less than 1/8 inch of the full length and width dimensions of the box and shall be fabricated of the same material as the box.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:44Z
Check the source to verify that this is the current version before use.

5.3 Unit loading. When specified (see 6.1), the product, packed as specified in 5.2.2 and 5.2.3, shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified. When unit loads are strapped, the strapping shall be limited to nonmetallic strapping.

5.4 Labeling. Each tray pack can shall be labeled in accordance with MIL-L-1497. In addition, the name of product shall be marked, stamping is permitted, on one 1001 by 200 side of the can. The labeling shall be legible when examined in accordance with 4.7 after submerging the can in boiling water for 30 minutes. Paper labels are not permitted. Cans shall show the following statements:

DO NOT HEAT

YIELD

Approximately eighteen (18) servings of 2/3 cup each.

5.5 Marking.

5.5.1 Shipping containers. In addition to any special marking required by the contract, shipping containers shall be marked in accordance with MIL-STD-129.

5.5.2 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078. In addition, the following precautionary markings in capital letters larger than other markings shall be included:

CAUTION: DO NOT STACK PALLETS IN TRANSIT OR MORE THAN TWO HIGH IN STORAGE

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this document.
b. When a first article is required (see 3.1, 4.3 and 6.3).
c. Provisions for approved preproduction samples (see 3.6.1 and 4.6.2).
d. Level of packaging and packing required (see 5.1 and 5.2).
e. Type and class of unit load, when unit loading is required (see 5.2.1 and 5.3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:44Z
Check the source to verify that this is the current version before use.

6.2 Appropriate level of pack. Based on conditions known or expected to be encountered during shipment, handling and storage of the specific item being procured, the procuring activity should select the appropriate level of pack in accordance with the criteria established in the AR 700-15/NAVSUPINST 4030.28/AFR 71-6/MCO 4030.33A/DLAR 4145.7.

6.3 First article inspection. When a first article is required it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should include specific instructions in all procurement documents regarding arrangements for selection, inspection, and approval of the first article.

Preparing activity:

Army - GL

Project No. 8940-A447

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:44Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY
US Army Natick Research and Development Center
Natick, Massachusetts 01760-5014

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300
STRNC-ES

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
US Army Natick Research and Development Center
ATTN: STRNC-ES
Natick, MA 01760-5014

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:44Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
EE-M-002206(Army-GL)	Macaroni Salad, Thermostabilized, Tray Pack

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

NATICK OP, 1 Apr 84

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:44Z
Check the source to verify that this is the current version before use.